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RESEARCH ARTICLE

Sustainable Supply Chain and Organisational Resilience: Evidence From China

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ABSTRACT

In the context of globalisation and resource constraints, embracing sustainable development has become a global consensus and an essential strategy for firms seeking to enhance their adaptability and resilience, with the sustainable supply chain playing a crucial role in this process. This study concentrates on Chinese listed firms from 2009 to 2022, investigating the impact of the sustainable supply chain level, formed by the ESG performance of other firms in the supply chain network, on organisational resilience. The findings indicate a positive correlation between the degree of sustainability of the supply chain network in which a firm is embedded and organisational resilience, with stronger risk resistance and adaptive abilities enabling firms to more effectively leverage sustainable supply chains to enhance their resilience. Heterogeneity analysis reveals a significant impact on non-dual-role and state-owned firms, and in regions with more favourable market environments. This study explores the transmission effects of ESG performance in the supply chain network, examining external factors affecting organisational resilience. These findings provide valuable insights for firms in emerging economies, aiming to enhance their sustainable development practices.

JEL Classification: L14, G34

1 | Introduction

Faced with multifaceted challenges such as the COVID-19 pandemic, geopolitical tensions and extreme events, the vulnerability and uncertainty in global business environments highlight the urgent need for resilience in supply chain management (Dai et al. 2021). Resilience refers to a firm's ability to withstand, adapt, recover and maintain operations amidst various challenges (Ortiz-de-Mandojana and Bansal 2016). As global supply chains become more complex and specialised, optimising supply chains is critical for enhancing corporate resilience (Negri et al. 2021).

Global supply chains are moving towards sustainability, driven by regulatory changes that emphasise improvements in environmental, social and governance (ESG) standards (Tseng et al. 2019).[1] At the core of these shifts is the integration of sustainability across the entire supply chain, which strengthens the control of environmental risks and social responsibilities through the implementation of ESG standards (Dubey et al. 2019; Starks 2009). Stakeholder theory emphasises the importance of effectively managing relationships within supply chains to promote ESG goals (Freudenreich et al. 2020). Strong ESG performance not only enhances a firm's market reputation but also directly influences the support of customers, suppliers and investors for sustainable practices within interdependent and interconnected supply chain networks (Rajesh 2020).

However, existing studies predominantly analyse relationships using a linear 'supplier-firm-customer' model (Truong

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et al. 2017), focusing on direct interactions between individual actors in terms of ESG performance (Tang et al. 2023). For instance, research shows that strong ESG practices by customers can extend upstream in the supply chain, positively impacting suppliers' ESG performance (Dai et al. 2021). Nevertheless, supply chain relationships should be viewed as part of a more complex network, where multiple interdependencies exist among suppliers, customers and other stakeholders (Choi and Krause 2006). Changes within one entity in this network can trigger corresponding adjustments in others (Yan et al. 2015). In this context, when customers implement sustainable procurement policies, they not only require their direct suppliers to adopt higher environmental and social standards, but also indirectly influence the practices of their suppliers' suppliers or other customers (Alghababsheh and Gallear 2021). This process strengthens sustainable practices across the entire supply chain network.

From a holistic network perspective, a sustainable supply chain is collectively shaped by the ESG performance of all partners embedded within the network (Chen and Xie 2022). By leveraging these resources, firms can significantly contribute to enhancing long-term sustainability and organisational resilience (Hillman et al. 2009).

Using customer and supplier data disclosed by Chinese listed firms from 2009 to 2022, this study constructs a unique annual supply chain network for each firm. Specifically, each focal firm establishes connections with multiple customers and suppliers, including both directly connected partners and indirectly related firms. Based on this data, the hierarchical structure of each firm's network is quantified, revealing the impact of both direct and indirect connections. Given that influence within supply chain networks exhibits a distance decay effect (Hearnshaw and Wilson 2013), this study applies Dijkstra's algorithm, using the ESG scores of other firms and their distance from the core firm as weights, to calculate the sustainability level of each core firm's supply chain.

The results indicate that the sustainability of the core firm's supply chain network significantly enhances its organisational resilience, with this effect being more pronounced for firms whose stronger ability to resist risks and adapt and adjust enables them to better leverage the benefits of a sustainable supply chain. Additionally, this study examines the heterogeneous effects of supply chain sustainability on firms' characteristics from three perspectives, finding that the impacts are more pronounced for non-dual-role firms, state-owned enterprises and regions with favourable market conditions.

This study offers several key contributions to the supply chain management literature: Firstly, it provides a network-level perspective on organisational resilience by examining how the sustainability of supply chain networks in which firms are embedded affects their ability to withstand and recover from disruptions. By integrating complex network theory with supply chain management, the study reveals that the ESG performance of network partners contributes to enhanced performance growth while mitigating financial volatility.

Secondly, it identified the moderating effects of risk resistance and the ability to adapt and adjust on the relationship between resilience and sustainability in supply chain networks. The ability to resist risks highlights a firm's capacity to withstand and mitigate short-term market fluctuations and unexpected disruptions, while simultaneously fostering long-term stability by improving preparedness for future uncertainties. In contrast, the ability to adapt and adjust focuses on long-term organisational resilience, emphasising a firm's capacity to strategically evolve and innovate in response to changing market dynamics. These findings deepen the theoretical understanding of supply chain research by elucidating the specific moderating roles through which sustainable supply chain networks enhance organisational resilience.

2 | Literature Review and Development of Hypotheses

2.1 | Sustainable Supply Chain Network and Organisational Resilience

Organisational resilience refers to an organisation's capacity to analyse and respond to a wide range of strategic opportunities and demands in a continuously changing environment (Grøgaard et al. 2022). It encompasses the ability to withstand shocks or enhance functionality during adversity (Do et al. 2022), and to adapt and transform under prolonged stress (DesJardine et al. 2019). In practice, organisational resilience is often viewed as a dual structure, characterised by high-performance growth and low financial fluctuations (Ortiz-de-Mandojana and Bansal 2016).

Many studies have confirmed that integrating ESG objectives into business strategies enhances long-term performance and risk management (Lee et al. 2022). First, the complexity of supply chains calls for effective sustainability management across different organisational levels (Lintukangas et al. 2023). When firms within a supply chain network demonstrate strong ESG performance, it indicates that the entire network can offer greater stability, mitigating the impact of systemic risks such as environmental regulations and social issues on individual firms. For the core firm, this stability ensures smooth production processes, reduces market losses due to supply chain disruptions and enhances market performance (Aguilera et al. 2021). Simultaneously, it helps reduce the transmission of financial volatility, as a key supplier with strong ESG performance can better protect the core firm's cash inflows and alleviate financial pressure during supply chain disruptions (Hendricks and Singhal 2005).

Second, sustainable supply chain networks indirectly improve the core firm's brand image and financing capacity (Lins et al. 2017). On one hand, the market's perception of socially responsible firms is influenced by the ESG performance of their suppliers or customers (Tang et al. 2023). This encourages investors to support firms that collaborate with others demonstrating strong ESG performance, thereby boosting their market competitiveness and reducing market volatility. On the other hand, financial institutions are increasingly considering the overall

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sustainability of a firm's supply chain when evaluating its creditworthiness (Jia et al. 2020). The partnerships formed serve as credit endorsements for the core firm, signalling stability to financial institutions. This, in turn, helps the core firm reduce financing costs and secure more favourable financing terms (Cao et al. 2021).

Finally, sustainable supply chain networks enhance information sharing and risk management between firms. Shared development goals foster trust, reducing information search and contract enforcement costs (Patatoukas 2012). Partners are more willing to signal current or potential changes in the supply chain environment, enabling joint identification and management of risks (Xie et al. 2022). This synergy reduces operational uncertainty and builds financial redundancy through stable cash flows and efficient asset utilisation, strengthening the firm's ability to withstand risks (Zhang et al. 2020). Based on these considerations, this study proposes the following hypothesis:

Hypothesis 1. *The sustainability of the supply chain network positively impacts organisational resilience.*

2.2 | The Role of the Ability to Resist Risks

The ability to resist risks refers to a firm's capacity to effectively identify, respond to, and recover from external shocks or internal crises, demonstrating how the firm utilises its existing resource reserves, risk prevention mechanisms, and emergency response systems to effectively manage crises (Dong et al. 2016). Firms with strong risk resistance can maintain operational stability despite external disruptions (Liang and Li 2024), which enables them to better capitalise on the benefits of sustainable supply chain networks, moderating the impact of sustainability on resilience.

Firms with a strong ability to resist risk play a pivotal role in amplifying the long-term benefits of sustainable supply chains, supported by their solid operational foundation and greater resource redundancy (Dong et al. 2016). When peer firms implement environmental initiatives—such as adopting green technologies (Liu et al. 2024) or sustainable procurement strategies (Truant et al. 2024)-these firms are better positioned to integrate such practices proactively. Drawing on their excess resources and organisational agility, they can more rapidly identify, comprehend and absorb ESG-related knowledge from the network, thereby enhancing the efficiency of technology adoption (Kong et al. 2020). These firms also place a high value on network stability and are capable of investing in long-term partnerships with sustainable supply chain stakeholders. This investment ensures a steady supply of raw material and smooth access to distribution channels. The resulting environmental collaboration generates business opportunities, attracts high-quality partners (Tang et al. 2023), and creates a positive feedback loop of value creation. Such dynamics facilitate the transition from transactional relationships to strategic alliances (Kotabe et al. 2003), thereby supporting sustained corporate growth.

At the same time, firms with a stronger ability to resist risk are also better positioned to realise the short-term benefits of sustainable supply chains. Owing to their financial flexibility, they can swiftly activate backup suppliers or draw on buffer inventories in response to minor or sporadic disruptions within the network (Tian et al. 2022). They can also leverage robust tangible resources to enable physical adaptability, such as shifting production lines or rerouting logistics (Dong et al. 2016). These capabilities help minimise the short-term impact of disruptions on production schedules and delivery, thereby avoiding sudden cost surges (Pankratz and Schiller 2024). In addition, such firms are more adept at insulating themselves from reputational spillovers within the network. For instance, through transparent supply chain traceability, they can demonstrate responsible management practices and avoid being implicated in partners' misconduct. This not only helps stabilise stock prices and sales but also prevents sharp short-term declines in performance. Therefore, we propose the following hypothesis:

Hypothesis 2. *When the firm is more resilient to risks, the positive correlation between the sustainability of the supply chain network and organisational resilience is more pronounced.*

2.3 | The Role of the Ability to Adapt and Adjust

The ability to adapt and adjust refers to a firm's capacity to maintain competitiveness and ensure long-term survival through continuous strategic adjustments and resource optimisation in a dynamic environment (Liang and Li 2024). This ability involves not only responding to short-term changes but also constitutes an ongoing process (Pike et al. 2010), where the firm proactively adjusts its strategy and resource allocation based on external shifts and historical context to address future challenges.

A strong ability to adapt and adjust enables firms to better capture and respond to the sustainability signals embedded within their supply chain networks. In the environmental dimension, peer firms' efforts to optimise resource use (Kong et al. 2020), which accelerate industry-ide green transitions, can be recognised early by firms with strong adaptability, and these firms can adjust their production processes and raw material procurement channels to meet the growing market demand for eco-friendly products, while also exploring new market opportunities arising from green development, thus positioning themselves for sustained long-term performance (Asgharian et al. 2024). On the social front, peer firms enhance the social reputation and image of the supply chain, and firms with strong adaptability, leveraging this positive social atmosphere, can support innovation and competitiveness, and through long-term engagement, further enhancing their brand image and social recognition, which ultimately leads to more business opportunities and long-term customer loyalty, thus achieving sustainable growth (Tao et al. 2023). In the field of corporate governance, the improved governance structures and transparent decision-making mechanisms of peer firms set an industry standard, with firms possessing strong adaptability able to rapidly absorb advanced practices from peer firms through long-term observation and learning, ensuring their competitiveness remains strong in the face of fierce market competition (Ersahin et al. 2024).

Short-term fluctuations, typically arising from sudden and unpredictable events, require firms to quickly mobilise resources and capabilities to maintain operational stability (Verreynne

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et al. 2023), and unlike long-term growth, responding to short-term fluctuations demands rapid adjustments, with adaptability and adjustment focusing on strategic and operational changes based on environmental shifts, but these adjustments, requiring time and resources, are harder to implement in the short term, making firms more likely to adopt temporary measures rather than comprehensive strategies when facing crises (Ersahin et al. 2024; Pike et al. 2010). While adaptability certainly plays a role in short-term responses, it proves more effective in long-term, relatively stable changes, where gradual adjustments to strategy and operations foster sustainable development. Therefore, we propose the following hypothesis:

Hypothesis 3. *When the firm possesses a strong ability to adapt and adjust, the positive correlation between the sustainability of the supply chain network and organisational resilience is more pronounced.*

3 | Data and Research Design

3.1 | Data

Since 2009, the China Securities Regulatory Commission has been encouraging listed firms to disclose detailed information, including the names or sales amounts of their top five customers (Li et al. 2022). This paper utilises A-share listed firms on the Shenzhen and Shanghai Stock Exchanges from 2009 to 2022 as research samples and processes the data as follows: (1) Firms in the financial industry, along with ST and *ST firms, are excluded. (2) Firms with the missing key variable data are excluded from the dataset. (3) Major continuous variables are subjected to 1% level winsorization, ultimately yielding 4298 firm-year samples. ESG rating indicators are sourced from Sino-Securities Index Information Service Co. Ltd. (Li et al. 2023), while both supply chain network relations indicators and financial data are obtained from the CSMAR database.

3.2 | Dependent Variables

Organisational resilience is characterised by its multilevel, multidimensional and dynamic nature, making direct measurement often subjective. Given this subjectivity, this study employs an indirect assessment method. Organisational resilience manifests through two primary dimensions in organisational contexts: First, it is demonstrated through adaptive growth capacity, specifically, a firm's ability to sustain performance growth when confronting external shocks or environmental uncertainties. Second, resilience is evidenced through operational stability, whereby resilient organisations display superior capabilities in risk anticipation, real-time environmental adaptation and rapid crisis recovery, consequently mitigating financial performance volatility.

This study, drawing from the research by Ortiz-de-Mandojana and Bansal (2016), conceptualises organisational resilience as a two-dimensional structure comprising growth and volatility. Considering that cumulative growth better reflects the long-term nature of growth capacity compared to annual growth, we measure a firm's long-term performance by using the cumulative increase in sales revenue over 3 years (DesJardine et al. 2019), which is defined as Growth. The financial volatility of a firm is assessed through the standard deviation of its monthly stock returns, which takes into account the reinvestment of cash dividends, which is defined as Vol. To ensure consistency in magnitude between the two dimensions, we apply reduction and amplification to the Growth and Vol variables respectively prior to regression analysis.

3.3 | Independent Variables

We utilise the ESG ratings from the Huazheng Index to assess corporate sustainable performance (Liu et al. 2023). The Huazheng ESG rating indicators integrate key elements of international ESG standards with developmental characteristics specific to China. The ESG ratings are classified into nine levels (AAA, AA, A, BBB, BB, B, CCC, CC and C) and converted into numerical variables from 9 to 1, with AAA assigned the highest value of 9 and C the lowest of 1, where a higher score reflects better ESG performance by the firm.

The network analysis method is employed to emphasise the indirect effects within supply chain networks, where a firm's behaviour is influenced not only by its direct contacts but also by other nodes within the broader network. Based on the supply chain business relationships of A-share listed firms from 2009 to 2022, the study constructs an undirected supply chain network using Python. The Dijkstra algorithm is used to calculate the shortest trade paths between any two firms in the supply chain network (Bhatia et al. 2021). The minimum number of trade actions required to connect two firms in the network is defined as the network distance (Wu et al. 2020).

Considering that the mutual influence between firms in a supply chain diminishes as the shortest distance increases (Arora and Brintrup 2021), this paper limits such influences to firms within the same supply chain network and the same year, categorising these firms as 'network peers' (Ahern and Harford 2014). To evaluate the sustainability of the supply chain network, we use the Inverse Distance Weighting (IDW) method for a weighted assessment of the ESG performance of all firms in the same network cohort (Barbulescu et al. 2020). Once the supply chain network and the shortest distances are defined, the weight $w_{ij,t}^{\text{net}}$ of firm i in year t concerning firm j within the same cohort is calculated as follows:

$$w_{ij,t}^{\text{net}} = \frac{\frac{1}{\text{distance}_{ij,t}}}{\sum_{j \neq i} \frac{1}{\text{distance}_{ij,t}}} \tag{1}$$

In Equation (1), $\text{distance}_{ij,t}$ represents the shortest network distance between firms i and j within the year t, the numerator is the reciprocal of the shortest path distance from j to i, while the denominator is the sum of the reciprocals of the network distances for all cohort firms on the shortest paths within i's network relationships, such that the sum of all cohort firms' weights for firm i in that year equals 1.

According to the concept of supply chain network distance, the influence of peer firms within the network on a target

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firm diminishes as the distance between them increases; consequently, the impact on the sustainable performance of the target firm also decreases with increasing distance. Therefore, the core explanatory variable ESG_network is calculated as follows:

$$\text{ESG_network}_{i,t} = \sum_{j \neq i} w_{ij,t}^{\text{net}} \text{ESG}_{j,t} \tag{2}$$

where $\text{ESG}_{j,t}$ represents the ESG score of peer firm j in year t and $\text{ESG_network}_{i,t}$ represents the overall sustainability level of the supply chain network excluding the target firm itself, in which the target firm i is situated in year t. Each listed firm in the supply chain network serves as a target firm and also functions as a network peer firm for other targets, thereby influencing the supply chain's sustainability level for all firms during the sample period.

3.4 | Mechanism Variables

This paper references Liang and Li (2024) and constructs two separate capability measurement systems for a firm's ability to resist risks and ability to adapt and adjust. The ability to resist risks reflects the degree of operational stability a firm can maintain in response to external shocks, which is largely shaped by its resource endowments and risk mitigation capacity. Drawing on existing literature (Dong et al. 2016; Liang and Li 2024; Tian et al. 2022), this paper selects tangible resources, intangible resources, and risk prevention as the first-level indicators of risk resistance. The second-level indicators include fixed assets and market share under tangible resources; intangible assets and staff education level under intangible resources; and debt-to-assets ratio, fixed assets turnover, and current assets turnover under risk prevention.

The ability to adapt and adjust refers to a firm's capacity to continuously adjust its strategy and resource allocation over time through the cumulative effects of short-term changes in response to a dynamic environment, thereby maintaining competitiveness and ensuring long-term survival. Key factors promoting a firm's ability to adapt to environmental changes include resource acquisition, resource reallocation, compensation incentive mechanisms, and innovation in products and services (Feng and Chen 2022; Shan et al. 2021; Zhang et al. 2022). Based on these theoretical insights, this paper measures adaptive capability using four first-level indicators: flexible resource allocation, external resource acquisition, personnel compensation incentives and positive product or service optimisation. These are further reflected through specific second-level indicators: flexible resource allocation is assessed in terms of the variability of R&D, capital and advertising expenditures; external resource acquisition is represented by government support received; personnel compensation incentives focus on non-executive employee remuneration schemes; and positive product or service optimisation is captured through both exploratory and exploitative innovation activities.

The two-level indicator system is presented in Table 1.

Following the construction of the firm's ability to resist risks and ability to adapt and adjust, the next step is to calculate the composite evaluation indices for these two capabilities based on the entropy weight method.

TABLE 1 | Indicator system for measuring mechanism variables.

Variables	First-level indicators	Second-level indicators[a]
The ability to resist risks (RRA)	Tangible resources	Fixed assets
		Market share
	Intangible resources	Intangible assets
		Staff education level
	Risk prevention	Debt to assets ratio
		Fixed assets turnover
		Current assets turnover
The ability to adapt and adjust (AAA)	Flexible resource allocation	Coefficients of variation for the three main types of annual R&D, capital and advertising expenditures of the sample firms
	External resource acquisition	Amount of government subsidies
	Personnel compensation incentives	Employee compensation incentives
	Positive Product or service optimization	Exploratory innovation
		Exploitative innovation

[a]Second-level indicators are operationalised as follows: staff education level is measured by the proportion of employees holding a bachelor's degree or above relative to the total number of employees; the coefficients of variation for R&D, capital, and advertising expenditures are computed as the ratio of the standard deviation to the average value for each expenditure category; employee compensation incentives are measured by subtracting total executive compensation from the cash paid to and on behalf of employees, and dividing the result by the number of non-executive employees in post; based on the first four digits of the International Patent Classification (IPC) codes, exploratory innovation is the count of patents with codes absent in the past 5 years but appearing for the first time in the current year, while exploitative innovation is the count of patents filed by a firm in the current year with codes present in its portfolio over the past 5 years.

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3.5 | Control Variables

Following previous research by Ortiz-de-Mandojana and Bansal (2016) and Shiu and Yang (2017), we include control variables at the firm level, including ESG, INST, Size, Board, TobinQ, BM, Indep and Cashflow, to reflect corporate governance structures and financial soundness. ESG represents a quantitative assessment of the firm's overall performance in environmental, social, and governance, INST reflects the level of control and influence exerted by institutional investors on the firm's decision-making, Size measures firm size, Board and Indep capture corporate governance breadth and independence, TobinQ signifies investor confidence, BM offers insight into firm valuation, and Cashflow reflects cash generation efficiency.

The symbols and specific definitions of the main variables are shown in Table 2.

3.6 | Model Setting

Following the theoretical model's analysis of the factors influencing organisational resilience, to validate the impact of the sustainability of the supply chain network on different dimensions of organisational resilience, this study employs multivariate regression analysis. Drawing on the research of Ortiz-de-Mandojana and Bansal (2016), we construct two baseline panel data regression models:

$$\mathrm{Vol}_{it} = \alpha_0 + \alpha_1 \mathrm{ESG_network}_{it} + \sum_{h}^{N} \beta_h \mathrm{Controls}_{it} + \sum \mathrm{FixedEffect} + \varepsilon_{it} \tag{3}$$

$$\mathrm{Growth}_{it} = \alpha_0 + \alpha_1 \mathrm{ESG_network}_{it} + \sum_{h}^{N} \beta_h \mathrm{Controls}_{it} + \sum \mathrm{FixedEffect} + \varepsilon_{it} \tag{4}$$

where Vol_{it} represents the volatility of firm i at time t; Growth_{it} represents the growth; $\mathrm{ESG_network}_{it}$ denotes the main independent variable; the sum of $\beta_h \mathrm{Controls}_{it}$ represents the control variables; FixedEffect refers to year fixed effects and firm fixed effects. And we clustered the standard errors for all regression coefficients at the firm level.

4 | Empirical Findings

4.1 | Descriptive Statistics

The descriptive statistics presented in Table 3 provide a foundational analysis of the variables in our study, offering insights into their distribution and central tendencies. The ESG_network, with a sample size of 4298, exhibits a mean score of 4.243 and a standard deviation of 0.785, reflecting a moderate level of sustainability within the supply chain networks of the firms surveyed. The minimum score recorded is 1, while the maximum reaches 6, with a median value of 4.249. These figures

TABLE 2 | Variable definitions.

Variable name	Variable symbol	Definition
Dependent variables		
Volatility	Vol	Standard deviation of monthly stock returns considering reinvestment of cash dividends
Growth	Growth	Three-year cumulative sales revenue growth
Independent variables		
The sustainability of the supply chain network	ESG_network	IDW ESG performance of supply chain network peer firms
Control variables		
The ESG score of the firm	ESG	Quantification of Huazheng ESG ratings
Institutional investor shareholding ratio	INST	Total institutional investor holdings divided by the floating share capital
Firm size	Size	Natural logarithm of total annual assets
Number of directors	Board	Natural logarithm of board member count
Tobin's *Q*	TobinQ	(Market value of tradable shares + number of non-tradable shares × net asset value per share + book value of debt)/total assets
Book-to-market ratio	BM	Book value/total market value
Percentage of independent directors	Indep	Number of independent directors divided by the total directors
Cash flow ratio	Cashflow	Net cash flow from operations divided by total assets
Mechanism variables		
Ability to resist risks	RRA	Construct a two-level indicator system. Refer to Table 1 for details
Ability to adapt and adjust	AAA	Construct a two-level indicator system. Refer to Table 1 for details

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TABLE 3 | Descriptive statistics.

Variables	*N*	Mean	SD	Min.	Max.	Median
Vol	4298	12.27	5.216	1.008	38.82	11.32
Growth	4298	0.575	2.782	−64.12	22.51	0.065
ESG_network	4298	4.243	0.785	1.000	6.000	4.249
ESG	4298	4.221	1.054	1.000	6.000	4.000
INST	4298	0.421	0.244	0.000	0.923	0.430
Size	4298	22.66	1.625	18.63	27.71	22.44
Board	4298	2.172	0.205	0.693	2.708	2.197
TobinQ	4298	1.817	1.150	0.681	8.143	1.440
BM	4298	1.337	1.416	0.013	7.859	0.841
Indep	4298	0.372	0.054	0.222	0.571	0.333
Cashflow	4298	0.047	0.070	−0.463	0.239	0.046
RRA	3527	0.016	0.016	0.003	0.106	0.011
AAA	4280	0.049	0.038	0.001	0.176	0.047

indicate that, although a number of firms exhibit relatively high ESG performance, there is considerable variation within the data, suggesting that a significant portion of firms still have opportunities for improvement in their sustainability practices.

Growth spans a wide range between its minimum and maximum values, indicating a substantial disparity in the performance trajectories of firms. Volatility, capturing the variation in stock returns, similarly exhibits a broad range, with significant differences between the lowest and highest values. This suggests that while some firms demonstrate relatively stable stock returns, others face substantial fluctuations, reflecting varying degrees of market risk and instability encountered by firms in different sectors. Furthermore, the remaining variables also fall within reasonable value ranges, supporting the overall consistency and feasibility of the data.

4.2 | Baseline Regression Results

Table 4 presents the baseline regression results, focusing on the impact of ESG_network performance on organisational resilience, measured through volatility (Vol) and growth (Growth). Columns (1) and (2) incorporate the core explanatory variable, ESG_network, along with firm and year fixed effects, without any control variables. Columns (3) and (4) incorporate additional control variables.

The regression findings suggest a significant negative correlation between ESG_network and volatility, while also showing a positive association with growth at the 1% significance level. In columns (1) and (2), the coefficients stand at −0.313 and 0.139, respectively, suggesting that a stronger sustainable supply chain network is associated with reduced volatility and enhanced growth. When control variables are included in columns (3) and (4), the coefficients for ESG_network remain significant, with values of −0.313 and 0.134, respectively. These findings support the hypothesis that firms with stronger sustainable supply chain networks enhance stability and foster growth.

We conduct discrete regression analyses in Table 5 for each aspect within the ESG_network to determine how each aspect contributes to organisational resilience (Karwowski and Raulinajtys-Grzybek 2021), and these analyses confirm and extend the baseline regression results.

The environmental dimension (E_network) significantly reduces volatility, indicating that firms with stronger environmental strategies are better equipped to manage financial risks, as these strategies mitigate the impacts of environmental disruptions, policy changes, and market fluctuations, thus enhancing long-term stability (Agliardi et al. 2023). Although a positive effect on growth is observed, it fails to reach statistical significance, likely due to the longer time required for returns from environmental investments to materialise into tangible financial performance. Social responsibility (S_network) significantly reduces volatility and increases growth, underscoring the importance of strong stakeholder relationships through social initiatives for both corporate expansion and stability (Karagiannopoulou et al. 2023). Governance (G_network) significantly reduces volatility and has a significant positive effect on growth, suggesting that strong governance frameworks are crucial for improving resilience (Ortas et al. 2014). In summary, these findings emphasise the complementary roles of each ESG dimension, with environmental strategies reducing financial risk, social initiatives fostering both stability and growth, and governance practices providing the necessary framework to drive long-term resilience.

4.3 | Robust Tests

4.3.1 | Instrumental Variable

To mitigate potential reverse causality issues in the baseline regression, we employ the instrumental variable (IV) approach. The characteristics of the target firm's supply chain network peers, including the attainment of both ISO14001 and ISO9001 certifications (ISO_network) and the location of peer

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TABLE 4 | Baseline regression results.

	(1)	(2)	(3)	(4)
Variables	**Vol**	**Growth**	**Vol**	**Growth**
ESG_network	−0.313** (0.123)	0.139*** (0.049)	−0.313*** (0.116)	0.134*** (0.048)
ESG			−0.217** (0.105)	0.084** (0.043)
INST			−2.029*** (0.689)	−0.064 (0.310)
Size			1.033*** (0.304)	0.873*** (0.176)
Board			−2.711*** (0.995)	0.673 (0.483)
TobinQ			1.234*** (0.170)	0.209*** (0.058)
BM			−1.164*** (0.140)	0.103 (0.119)
Indep			−4.539* (2.669)	2.938 (2.355)
Cashflow			−1.382 (1.383)	2.732*** (0.624)
Constant	16.622*** (0.626)	−0.320 (0.222)	1.162 (7.240)	−22.819*** (5.088)
Firm FE	YES	YES	YES	YES
Year FE	YES	YES	YES	YES
Observations	4298	4298	4298	4298
R^2	0.255	0.022	0.318	0.052

Note: This table reflects the impact of sustainable supply chain network on organisational resilience. All variables are defined in Table 1. Robust standard errors are clustered by firms and are reported in parentheses. All continuous variables were winsorised at 1% and 99% levels, and ***, ** and * indicate significances at the 1%, 5% and 10% levels (the number and presence of significance markers depend on the actual regression outcomes of the variables in the model).

TABLE 5 | Multidimensional regression results.

	(1)	(2)	(3)	(4)	(5)	(6)
Variables	**Vol**	**Vol**	**Vol**	**Growth**	**Growth**	**Growth**
E_network	−0.278*** (0.105)			0.037 (0.032)		
S_network		−0.334*** (0.111)			0.150*** (0.045)	
G_network			−0.220** (0.102)			0.118** (0.046)
Controls	YES	YES	YES	YES	YES	YES
Firm FE	YES	YES	YES	YES	YES	YES
Year FE	YES	YES	YES	YES	YES	YES
Observations	4298	4298	4298	4298	4298	4298
R^2	0.318	0.318	0.318	0.050	0.052	0.052

Note: This table reflects the heterogeneity results across different aspects of the sustainability of the supply chain network. All variables are defined in Table 1. Robust standard errors are clustered by firms and are reported in parentheses. All continuous variables were winsorised at 1% and 99% levels, and ***, ** and * indicate significances at the 1%, 5% and 10% levels (the number and presence of significance markers depend on the actual regression outcomes of the variables in the model).

firms in 'low-carbon pilot cities' (LCC_network), serve as instruments. ISO certifications signify a firm's commitment to environmental management and quality assurance, while the regulatory frameworks in low-carbon pilot cities likely incentivise enhanced sustainable practices. These factors contribute to increased efficiency and sustainability, impacting peer firms without directly influencing the organisational resilience of target firms, thereby meeting the essential criteria for instrumental variable selection. The construction method for these instrumental variables mirrors that used for the baseline regression's explanatory variables, employing the inverse network distance weighting approach.

Columns (1), (2) and (3) of Table 6 present the results of the IV-2SLS analysis. The instrumental variable tests passed both the Kleibergen-Paap rk LM test and the Hansen *J* test, indicating the chosen set of instrumental variables is valid and free from under-identification or over-identification problems. In the second-stage regression, the coefficient for the sustainability of the supply chain network remains statistically significant and

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TABLE 6 | Robust tests: IV and PSM.

	IV			PSM	
	(1)	(2)	(3)	(4)	(5)
Variables	**ESG_network**	**Vol**	**Growth**	**Vol**	**Growth**
ESG_network		−1.656*	0.619*	−0.270*	0.199***
		(0.979)	(0.340)	(0.159)	(0.077)
ISO_network	0.137***				
	(0.034)				
LCC_network	0.138***				
	(0.052)				
Controls	YES	YES	YES	YES	YES
Firm FE	YES	YES	YES	YES	YES
Year FE	YES	YES	YES	YES	YES
Observations	3652	3652	3652	2947	2947
R^2		0.285	0.029	0.353	0.077
Kleibergen-Paap rk LM	22.357***				
Hansen J (p)	0.2938				

Note: This table reflects the regression results after using the IV and PSM. ISO_network and LCC_network are two instruments. All other variables are defined in Table 1. Robust standard errors are clustered by firms and are reported in parentheses. All continuous variables were winsorised at 1% and 99% levels, and ***, ** and * indicate significances at the 1%, 5% and 10% levels (the number and presence of significance markers depend on the actual regression outcomes of the variables in the model).

positive. This suggests that the primary regression results are robust, even after accounting for potential sample selection biases.

4.3.2 | Propensity Scores Matching Test

We employ the Propensity Score Matching (PSM) method to minimise the interference of other factors. In the first step, we calculate the median ESG_network scores for each industry-year and subsequently generate the treatment variable (treat) based on these median values. Firms with ESG_network scores at or above the median are assigned to the treatment group (treat = 1), while those below the median are assigned to the control group (treat = 0). Next, we estimate the propensity scores for treatment assignment using a logistic regression model and perform 1:1 nearest neighbour matching, applying a calliper of 0.05 to ensure that the propensity score differences between matched samples are confined within an acceptable range. All control variables were included to account for differences between the treatment and control groups. Finally, we assess the balance of the matched samples by examining kernel density plots. Figure 1 shows the propensity score distributions of the treatment and control groups before and after matching. The results reveal that, after matching, the distributions of propensity scores in the treatment and control groups almost entirely overlap, indicating a substantial reduction in selection bias and confirming the effectiveness of the matching procedure.

The matched regression results, shown in columns (4) and (5) of Table 6, indicate that a sustainable supply chain network has a significantly negative effect on financial volatility and a positive effect on performance growth. This demonstrates that the results remain robust even after controlling the impact of firm heterogeneity factors on organisational resilience.

4.3.3 | Heckman Two-Stage Model

To address potential sample selection bias due to the voluntary disclosure of supply chain information by listed firms, we adopt the Heckman two-stage method to mitigate its impact on our findings. In the first stage, we treat the decision to disclose supply chain information as the dependent variable (Disclose) and use industry-level disclosure averages as the instrumental variable, applying the ordered probit regression. This approach assumes that while the disclosure level of peers influences the decision to disclose, it does not directly affect organisational resilience.

Column (1) of Table 7 reports the first-stage regression results. In the second stage, we include IMR obtained from the first stage regression into the baseline model, as shown in columns (2) and (3) of Table 7. The results from these regressions indicate that, after adjusting for selection bias, the sustainability of supply chain networks continues to have a positive effect on the organisational resilience of core firms, further affirming the robustness of the primary findings in this study.

4.3.4 | Replacing the Organisational Resilience Measurement

To ensure the robustness of our analysis, we implement another measure of organisational resilience. Building upon the individual

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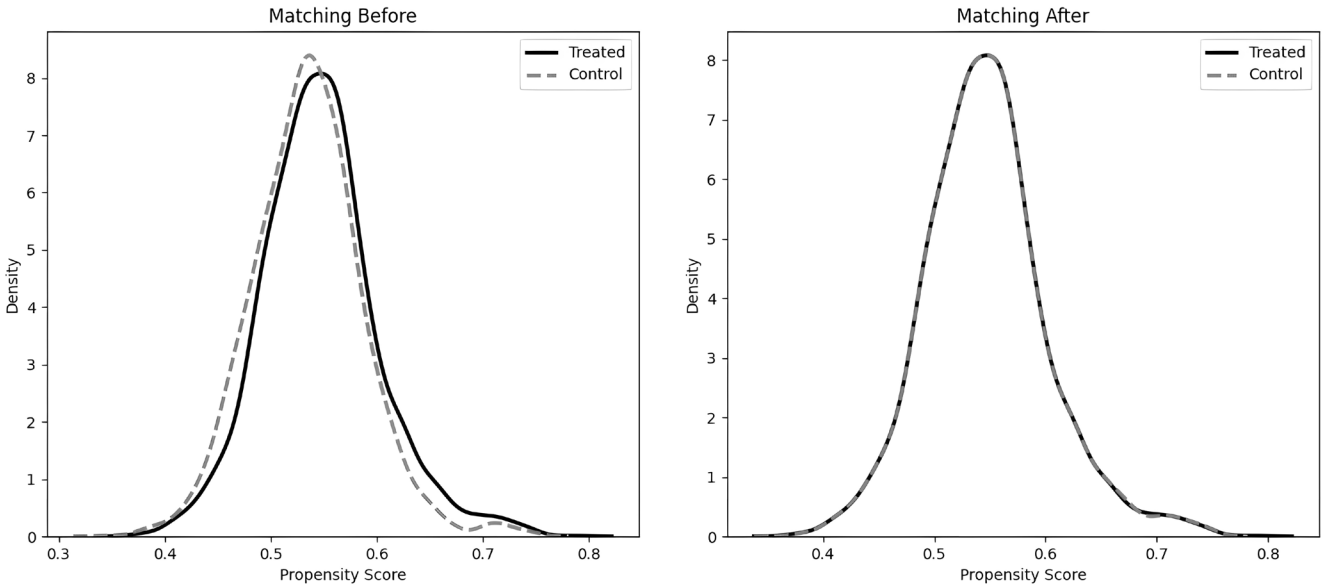

FIGURE 1 | Propensity score before and after matching.

TABLE 7 | Robust tests: Heckman two stage model and entropy method.

	Heckman two stage			Entropy method
	(1)	(2)	(3)	(4)
Variables	**Disclose**	**Vol**	**Growth**	**Resilience_Score**
Average_disclosure	4.417*** (25.554)			
ESG_network		−0.345*** (−2.991)	0.118** (2.576)	0.002*** (0.001)
IMR		−0.989 (−0.407)	−0.767 (−1.323)	
Controls	YES	YES	YES	YES
Firm FE	YES	YES	YES	YES
Year FE	YES	YES	YES	YES
Observations	41,366	4253	4253	4298
R^2	0.538	0.309	0.069	0.818

Note: This table reflects the regression results after using the Heckman two-stage model and the entropy method. Resilience_Score is a composite index reflecting overall resilience performance. IMR represents the inverse Mills ratios from the first stage regressions. All other variables are defined in Table 1. Robust standard errors are clustered by firms and are reported in parentheses. All continuous variables were winsorised at 1% and 99% levels, and ***, ** and * indicate significances at the 1%, 5% and 10% levels (the number and presence of significance markers depend on the actual regression outcomes of the variables in the model).

indicators of volatility and growth, we apply the entropy method to synthesise these two indicators into a composite index (Sun and Govind 2022), namely Resilience_Score, which reflects the overall resilience performance of a firm in a given year, with higher Resilience_Score indicating better resilience of the firm.

If ESG_network continues to have a significant impact on the Resilience_Score of this comprehensive measure, this will further confirm that the strong sustainable supply chain network of the firm will effectively enhance the resilience of the firm itself, helping the firm achieve long-term sustainable growth and staying stable amid uncertainty and shock. The result is shown in column (4) of Table 7, which supports the conclusions.

4.4 | Mechanism Tests

This study adopts the approach proposed by Rajan and Zingales (1996) to conduct a mechanism test, utilising an interaction model to explore how sustainable supply chain networks enhance the organisational resilience of core firms. If sustainable supply chain networks significantly strengthen the organisational resilience of firms with stronger risk resistance ability (ability to adapt and adjust), the interaction term between RRA (AAA) and ESG_network is expected to exhibit a significantly negative coefficient for the dependent variable Vol and a significantly positive coefficient for the dependent variable Growth.

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The regression results presented in columns (1) and (2) of Table 8 confirm this hypothesis. The positive relationship between ESG_network and organisational resilience is significantly strengthened in firms with higher levels of risk resistance, suggesting that risk resistance acts as a moderator in this relationship. This finding suggests that firms with higher levels of risk resistance are better positioned to leverage the sustainability of their supply chain networks to enhance their overall organisational resilience, highlighting the synergistic effects of sustainability and risk management in fostering long-term corporate resilience.

Based on the regression results from column (4) of Table 8, the interaction term shows a significant positive correlation with organisational growth. The stronger a firm's adaptability and adjustment capacity, the more pronounced the effect of a sustainable supply chain network on its growth. This can be attributed to the fact that firms with higher levels of adaptability are better able to leverage their sustainable supply chain networks to respond to changing market conditions, optimise their operations, and seize new growth opportunities, thereby enhancing their long-term growth trajectory and organisational resilience.

However, although this adaptability and adjustment capacity has a synergistic effect in promoting long-term growth, its significance in the short-term volatility regression model is not as pronounced, as indicated by the regression results in column (3) of Table 8. This can be attributed to the mismatch between the formative characteristics of a firm's adaptability and adjustment capacity, which is a long-term process (Pike et al. 2010) requiring continuous resource investment and experience accumulation, and the triggers of short-term fluctuations that are mostly caused by sudden external market shocks or unexpected events. These shocks, characterised by immediacy and unpredictability, exert impacts at a speed far exceeding the effective pace of the firm's adaptability and adjustment capacity, meaning that even firms with strong adaptability and adjustment capacity struggle to quickly activate the advantages of sustainable supply chains in the short term to buffer such shocks.

4.5 | Heterogeneity Analysis

4.5.1 | CEO Duality

CEO duality may influence corporate decision-making, thereby affecting supply chain selection and organisational resilience (Alves 2023). We classify firms into dual-role (dual) and separate-role (non-dual) firms by the combined roles of CEO and chairman to conduct a heterogeneity test.

The results in columns (1) to (4) of Table 9 indicate that firms with CEO duality show non-significant regression coefficients for the sustainability performance of their supply chain networks, while firms with separate CEO and chairman roles demonstrate notably positive coefficients at the 1% significance level. The separation of these roles enhances the positive impact of sustainable supply chain practices on long-term performance growth, financial risk reduction and organisational resilience. A decentralised governance structure may offer more advantages in harnessing the sustainable development benefits of supply chains, to boost organisational resilience and foster business growth (Mubeen et al. 2021).

4.5.2 | State Ownership

The impact of ESG_network on organisational resilience varies by ownership type, with a heterogeneity test conducted between state-owned firms (SOEs) and non-state-owned firms (Non-SOEs). As shown in columns (5) to (8) of Table 9, the results demonstrate differing effects of ESG_network on both volatility and growth for the two types of firms.

Firstly, in terms of growth, both SOEs and Non-SOEs exhibit a significant positive impact of ESG_network on organisational growth, where the effect in SOEs is stronger than that in Non-SOEs, indicating that SOEs can drive growth more robustly by

TABLE 8 | Mechanism tests.

	(1)	(2)	(3)	(4)
Variables	**Vol**	**Growth**	**Vol**	**Growth**
ESG_network	−0.345*** (0.128)	0.080** (0.036)	−0.281** (0.115)	0.127*** (0.045)
RRA	48.766** (21.436)	51.610*** (18.621)		
ESG_network × RRA	−11.752* (6.314)	17.225** (8.398)		
AAA			−5.258 (3.273)	−1.830 (3.836)
ESG_network × AAA			−4.895 (3.026)	4.357** (2.045)
Controls	YES	YES	YES	YES
Firm FE	YES	YES	YES	YES
Year FE	YES	YES	YES	YES
Observations	3527	3527	4280	4280
R^2	0.334	0.069	0.319	0.054

Note: This table reflects the regression results utilising supply chain stability and concentration as mediating effects. All other variables are defined in Table 1. Robust standard errors are clustered by firms and are reported in parentheses. All continuous variables were winsorised at 1% and 99% levels, and ***, ** and * indicate significances at the 1%, 5% and 10% levels (the number and presence of significance markers depend on the actual regression outcomes of the variables in the model).

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TABLE 9 | Heterogeneity analysis: CEO duality and state ownership.

	Non-dual		Dual		Non-SOEs		SOEs	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Variables	**Vol**	**Growth**	**Vol**	**Growth**	**Vol**	**Growth**	**Vol**	**Growth**
ESG_network	−0.343*** (0.131)	0.145*** (0.052)	−0.444 (0.290)	0.016 (0.091)	−0.270 (0.185)	0.090** (0.041)	−0.275* (0.146)	0.174** (0.086)
Controls	YES	YES	YES	YES	YES	YES	YES	YES
Firm FE	YES	YES	YES	YES	YES	YES	YES	YES
Year FE	YES	YES	YES	YES	YES	YES	YES	YES
Observations	3318	3318	980	980	2320	2320	1978	1978
R^2	0.312	0.055	0.414	0.078	0.330	0.122	0.336	0.065

Note: This table reflects the regression results for CEO duality and state ownership. All other variables are defined in Table 1. Robust standard errors are clustered by firms and are reported in parentheses. All continuous variables were winsorised at 1% and 99% levels, and ***, ** and * indicate significances at the 1%, 5% and 10% levels (the number and presence of significance markers depend on the actual regression outcomes of the variables in the model).

TABLE 10 | Heterogeneity analysis: External environment.

	West		Mid		East	
	(1)	(2)	(3)	(4)	(5)	(6)
Variables	**Vol**	**Growth**	**Vol**	**Growth**	**Vol**	**Growth**
ESG_network	−0.124 (0.217)	0.051 (0.096)	−0.464 (0.307)	0.128 (0.100)	−0.280* (0.155)	0.167** (0.067)
Controls	YES	YES	YES	YES	YES	YES
Firm FE	YES	YES	YES	YES	YES	YES
Year FE	YES	YES	YES	YES	YES	YES
Observations	1008	1008	603	603	2686	2686
R^2	0.292	0.099	0.371	0.186	0.341	0.054

Note: This table reflects the regression results for the external environment. All other variables are defined in Table 1. Robust standard errors are clustered by firms and are reported in parentheses. All continuous variables were winsorised at 1% and 99% levels, and ***, ** and * indicate significances at the 1%, 5% and 10% levels (the number and presence of significance markers depend on the actual regression outcomes of the variables in the model).

improving ESG performance within their supply chain network while Non-SOEs also benefit from such enhancements to foster growth, with the stronger impact in SOEs possibly related to their resource advantages and policy support in promoting ESG practices (Liao et al. 2024).

Secondly, in terms of volatility, ESG_network exerts a significant volatility-reducing effect on SOEs, which is not observed in Non-SOEs, suggesting that optimising ESG performance within the supply chain network helps reduce volatility more notably for SOEs, as their relatively standardised operational mechanisms allow ESG_network advantages to be more effectively translated into risk mitigation effects, while for Non-SOEs, the non-significant effect may reflect the influence of more complex market factors that weaken the immediate impact of ESG_network on volatility reduction (Chen et al. 2022).

4.5.3 | External Environment

Regional variations in economic environments, policy support, and market demands may lead to divergences in the effectiveness of sustainable supply chain network practices and resilience. We categorise firms based on their locations into Eastern, Central and Western regions,[2] presenting results in Table 10. The findings show that the influence of sustainability on organisational resilience is significantly stronger in the Eastern region, while the effects are not statistically significant in either the Central or Western regions.

The differences in the implementation of sustainable development practices among peer firms within supply chain networks are likely attributable to the Eastern region's advantage in terms of economic development, policy support, and sustainable awareness compared to the Western and Central regions, as the East typically features a more developed market environment, coupled with higher levels of sustainable implementation. These factors synergistically enable firms in the Eastern region to more effectively translate network-wide sustainable initiatives into enhanced organisational resilience. Thus, the marked regional differences indicate that Eastern firms are better positioned to leverage the sustainable supply chain network to augment their adaptability to market fluctuations and bolster long-term development.

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5 | Conclusion

This study explores how the sustainable supply chain network affects a firm's organisational resilience. A key feature of this research is the introduction of the concept of indirect effects of peer networks, which extend beyond directly connected suppliers and customers, into sustainable supply chain research, thereby broadening the scope of resilience to encompass the impact of the broader network rather than solely focusing on direct relationships. The empirical findings demonstrate that a robust and sustainable supply chain network significantly contributes to organisational resilience by mitigating short-term fluctuations and fostering long-term growth.

The mechanism analysis indicates that the firm with a strong ability to resist risks experiences a more significant impact of a sustainable supply chain on its organisational resilience, while the firm with a strong ability to adapt and adjust is more profoundly influenced by the sustainable supply chain in terms of long-term growth as an aspect of organisational resilience. Additionally, the study uncovers important nuances regarding the varying impacts of sustainable supply chain networks across different types of firms. Specifically for this heterogeneity, the positive effect is significantly stronger in firms with separated CEO and chairman roles; for state-owned firms, the effect of ESG_network on growth is stronger compared to non-state-owned firms, and its role in reducing financial volatility is more pronounced than that in non-state-owned firms; and the impact is most evident among firms located in the Eastern region of China.

Emerging economies are often characterised by underdeveloped regulatory frameworks and market inefficiencies, which hinder firms' ability to internalise the positive spillovers of sustainable supply chain practices. In response, policymakers can adopt three targeted measures: (1) Develop a tiered sustainability certification system for firms adopting ESG-compliant supply chain practices, aligned with international standards, with differentiated institutional benefits to encourage broad participation and strengthen the resilience and global integration of regional supply chain ecosystems; (2) Promote capability-building initiatives that enhance firms' ability to resist risks and adapt, enabling them to leverage sustainable supply chain networks for organisational resilience, while encouraging managers to engage partners aligned with sustainability goals and collaborate on improving environmental and social performance across the supply chain to mitigate market fluctuation risks and enhance operational stability; (3) Establish sustainability governance frameworks tailored to firm-level heterogeneity, with the goal of facilitating the internalisation of the positive environmental externalities generated by sustainable supply chain practices through a combination of regulatory incentives and market-based mechanisms.

While the study offers valuable insights into the relationship between sustainable supply chain networks and organisational resilience, its scope is limited to emerging economies, which may constrain the generalisability of the findings to advanced economies. Future research could address this by examining cross-national variations in the impact of sustainable supply chain networks on organisational resilience across different institutional and economic contexts.

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Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

Endnotes

[1] For example, the European Union's Corporate Sustainability Reporting Directive (2022) and Corporate Sustainability Due Diligence Directive (2023) have strengthened supply chain oversight and disclosure requirements. The Hong Kong Stock Exchange's ESG Reporting Guidelines emphasise green procurement and ESG risk management in supply chain practices.

[2] The regional classification follows Shen et al. (2021), which categorises firms into Eastern, Central, and Western China based on their geographic locations.

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